

Building-age transfer with scarce labels

Madrid → Amsterdam

We classify four building-construction eras from 30 m Landsat pixels. Coordinate-RF combines a Madrid-learned prior with a target-support random forest, whitened spectral features and spatial context. On the frozen Amsterdam random-pixel protocol it reaches 0.749299 macro-F1 at 200 labels per class. The method uses the full unlabelled target pool and an additional development label bank; spatial extrapolation remains a limitation.

0.749299

Locked macro-F1

200 labels/class

200 paired episodes

A useful urban signal, with imperfect labels

Why age?

Retrofit planning
Energy-policy evidence
Incomplete records

What data?

Six spectral bands
1984–2025 imagery
30 m mixed pixels

What task?

Four construction eras
Madrid source city
Amsterdam target city

76,263 source pixels · 25,992 target pixels · 60 spectral features

Transfer must work when labels are scarce

Cross-city shift

Spectral and spatial patterns differ.
Source knowledge needs adaptation.

Few-label adaptation

5 / 25 / 50 / 100 / 200 labels per class.
Only designated support labels fit the local RF.

25 labels/class = 100 target support labels

The additional 800-label development bank is separate research supervision.

One source prior, one local adaptation branch

Madrid source branch

Contextual spectral features
Covariance alignment + refinement
Frozen source probability prior



Amsterdam local branch

Full-pool adaptive whitening
Two standardized coordinates
Support-only 200-tree RF



Blend probabilities → nine-neighbour smoothing → classes

The contribution is the combination

Source reuse

Source probability
weight:

$$20 / (20 + 4b)$$

Less weight as support
grows.

Local geometry

Adaptive whitening

Spectral + coordinate
inputs

Balanced random forest

Spatial context

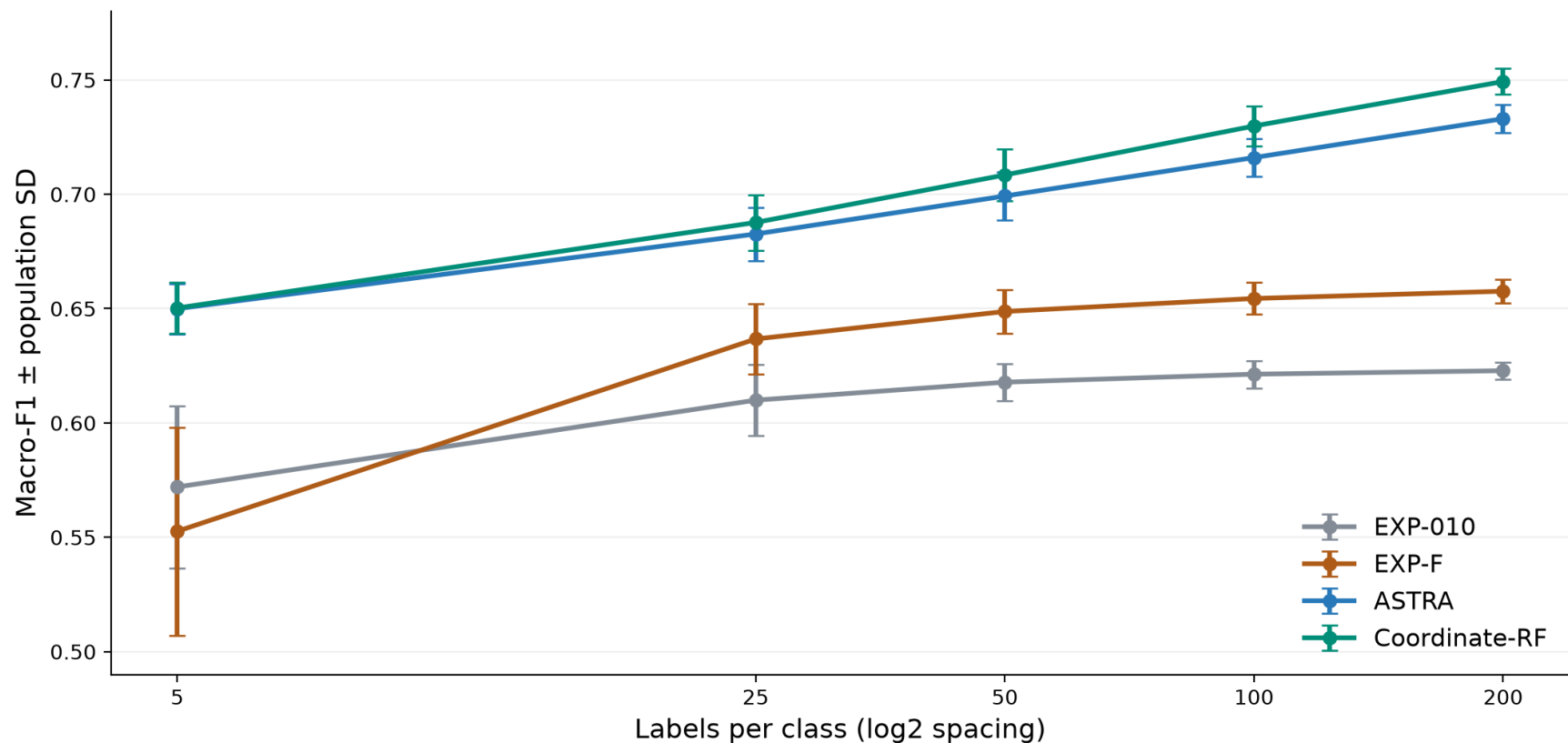
Label-free smoothing

Nine spatial neighbours

Full-pool access required

b = labels/class. Components interact; ablation bars are not additive causal effects.

Performance improves as target support grows



25 / class

0.687664

200 / class

0.749299

SD is episode spread, not a confidence interval.

A strong fixed-city result—not universal superiority

200 labels/class

Coordinate-RF: 0.749299 ± 0.005716

Matched ASTRA: 0.733037

Mean gain: +0.016262

What the evidence says

200/200 paired wins at 200/class

Five-shot superiority is inconclusive

One shared target-city population

The 800-label development bank and current support are excluded from final queries.

Spatial extrapolation reveals the trade-off

Four-direction mean

Coordinate-RF: 0.694216

Matched ASTRA: 0.696672

Worst-direction mean

Coordinate-RF: 0.644816

Matched ASTRA: 0.658183

Coordinates help within this pool; they do not establish transfer to new geography.

Frozen evidence, explicit reproducibility boundaries

What is preserved

Model + configuration + ordered inputs
Notebook outputs + result tables
Manifest and SHA-256 checksums

What is not claimed

Fresh training or runtime replay here
A self-contained raw-data rebuild
Organizer-held-out validation

Safe audit: `python -B scripts/validate_release.py`

A clear result—and a clear next test

Coordinate-RF · macro-F1 0.749299

200 labels/class · fixed-city random-pixel evaluation

Before competition upload

Confirm full-pool, coordinate and additional-supervision eligibility.

Next research—not run

Independent-city / spatial validation
Portable inference for new target pools